

Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.



Executive data report

17%

Global traffic growth

Year-over-year increase
in 2024

225

Major disruptions

Observed globally during
the year

6.5%

Traffic mitigated

Share of all processed
traffic

13%

Post-quantum TLS

Approx. TLS 1.3 traffic
adoption

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

Global Internet traffic

Year-over-year growth as reliance on Internet services continued to expand.

3.3x

Starlink traffic surge

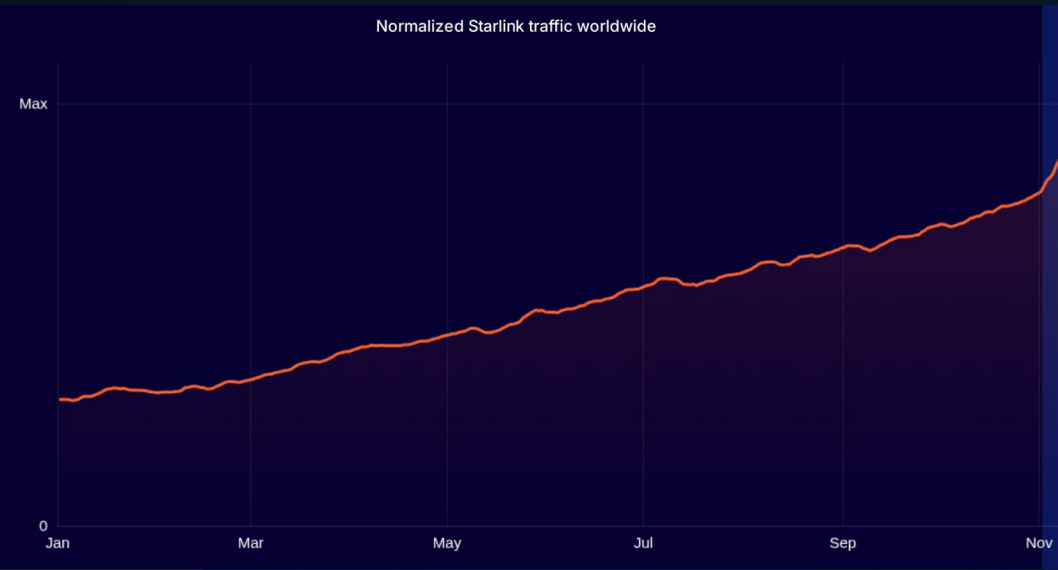
SpaceX Starlink traffic grew dramatically while expanding satellite connectivity.

41%

Mobile traffic

Global traffic came from mobile devices.

Trend line starts mid-January to normalize for post-holiday activity; Starlink is scaled relative to peak observed request volume.



Normalized Starlink traffic worldwide showing 3.3x growth in 2024

Executive readout

- The macro traffic baseline continues rising, reinforcing capacity, performance, and resilience planning needs.
- Satellite Internet is becoming a measurable growth vector for underserved and hard-to-reach regions.
- Mobile's 41% share keeps edge optimization and last-mile variability central to user experience.

Google Leads Top 10 Services as Generative AI Rises

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Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Top 10 most popular Internet services worldwide

Signal: Google held #1 for the second consecutive year; generative AI continued its rapid rise throughout 2024 while the Metaverse receded into the background.

Ranking from Cloudflare 1.1.1.1 DNS resolver

#1 Google

#6 Microsoft

#2 Facebook

#7 Instagram

#3 Apple

#8 YouTube

#4 TikTok

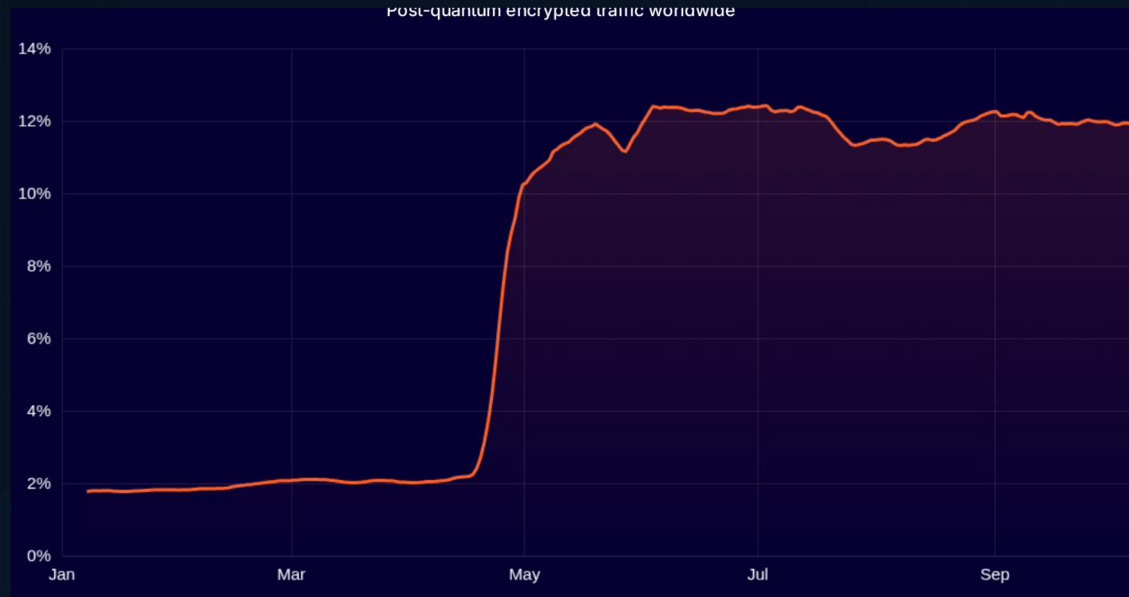
#9 Amazon

#5 AWS

#10 WhatsApp

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

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2%

Early 2024

Starting point for post-quantum TLS 1.3 traffic.

13%

Rapid climb

Approximately 12–13% after Chrome 124 default enablement.

Trigger point

Google Chrome 124 enabled post-quantum key agreement by default on April 17. Mozilla Firefox started rolling out post-quantum by default; Apple Safari began initial testing.

Post-quantum encrypted traffic worldwide after Chrome 124 default enablement

21%

HTTP/3 adoption

Runs on QUIC with faster connections and encryption by default.

Go

Top API client

Approximately 12% of automated API requests.

33%

iOS traffic

iOS devices generated one-third of traffic.

89%

Google search

Share of search engine referral traffic.

225 Major Disruptions: Government Shutdowns Lead Outage Causes

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225

Major disruptions

Observed globally in 2024.

116+

Government Directed

Leading recorded cause.

41

Cable Cut

Second-highest cause.

16

Unknown

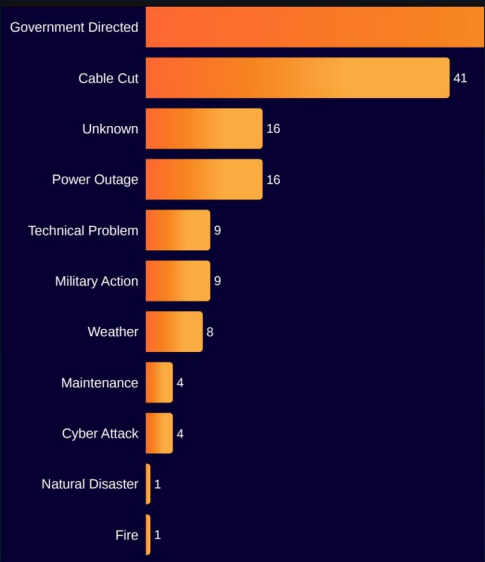
Recorded incidents.

16

Power Outage

Recorded incidents.

Cause breakdown



Causes of major Internet disruptions observed in 2024

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

All traffic mitigated

Scale of ongoing threats across Cloudflare-processed traffic.

68%

Bot concentration

Global bot traffic originating from top 10 countries.

4.3%

Emails malicious

Share of processed emails identified as malicious.

43%

Deceptive links

Among malicious emails, share containing deceptive links.

Most-attacked sector

Gambling and Games

The sector was the most attacked industry category in Cloudflare's 2024 observations.

Known vulnerabilities persist

Log4j (CVE-2021-44228), first disclosed in December 2021, continues to be actively targeted more than two years later.

Routing and email signals

- RPKI valid routes increased by 6.4%.
- The .bar top-level domain originated the largest share of malicious and spam email observed.
- Email link deception remains a high-volume abuse pattern.

Five Takeaways Shaping the 2025 Internet Outlook

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1

Traffic growth remains structural

Global Internet traffic grew 17% year-over-year, with Starlink traffic surging 3.3x as satellite connectivity expanded to underserved regions.

2

Post-quantum momentum is real

Adoption leapt from approximately 2% to 13% of TLS 1.3 traffic following Chrome 124 default enablement in April 2024.

3

Connectivity fragility is geopolitical

Government-directed shutdowns remained the leading cause of the 225 major Internet disruptions observed.

4

Threat pressure is persistent

6.5% of all traffic was mitigated; Gambling/Games was most targeted, and Log4j exploitation persists years after disclosure.

5

Developer ecosystems keep shifting

Go became the most popular API client language, capturing roughly 12% of automated API traffic.

2025 planning lens: capacity, cryptography, resilience, security, and developer-platform strategy